

DATA SCIENCE REVIEW – 2

Handwritten Text Recognition using Deep Learning

Roll Number	Name
2420030231	Sneha Bomma
2420030233	Kola Hitheshini
2420030445	Lalitha mahisri Mallepudi

Department of Computer Science and Engineering



KONERU LAKSHMAIAH EDUCATION FOUNDATION

Introduction

- Handwritten text recognition is an important application of deep learning and computer vision that enables computers to recognize and convert handwritten characters into digital text. It helps automate the analysis of handwritten information and reduces the need for manual text entry.
- This project focuses on developing a deep learning model using the EMNIST dataset, which contains handwritten letters and digits. A Convolutional Neural Network (CNN) learns visual patterns such as strokes, edges, shapes, and character structures to classify handwritten characters.

Problem Statement

- Handwritten characters can vary significantly in their strokes, shapes, size, and writing style, making automatic recognition a challenging task. Converting handwritten information into digital text manually can also be time-consuming when large amounts of data are involved.
- Therefore, there is a need for an automated system that can recognize handwritten characters accurately and convert them into digital text. This project proposes a CNN-based deep learning model trained on EMNIST to perform handwritten character recognition.

Objectives

- The primary objective of this project is to develop a deep learning-based system for recognizing handwritten characters using the EMNIST dataset. The model is designed to learn important visual patterns from labelled handwritten character images.
- The project aims to perform data preprocessing and image normalization, develop and train a CNN, validate the model, and evaluate its performance using accuracy, loss, precision, recall, and confusion matrix. The trained model is then used to recognize handwritten characters and convert them into digital text.

Literature Survey

- Handwritten character recognition has been explored using traditional machine learning and neural network techniques. Earlier approaches often depended on manually designed features to represent strokes, edges, and shapes before performing classification.
- Deep learning approaches, particularly Convolutional Neural Networks (CNNs), can automatically extract relevant features from handwritten images. CNN-based methods are therefore well suited for learning character structures and classifying handwritten letters and digits.

Literature Gap

- Existing handwritten character recognition approaches demonstrate the effectiveness of machine learning and deep learning techniques, but a complete and simple workflow from image preprocessing to digital text conversion is not always presented as a single system.
- This project focuses on integrating preprocessing, image normalization, CNN-based feature extraction, training, validation, performance evaluation, and character prediction into one basic handwritten text recognition pipeline.

Flow of the Project

- The project begins with loading the EMNIST dataset containing labelled handwritten letters and digits. The character images are preprocessed and normalized before being provided to the CNN model, which automatically extracts relevant visual features such as strokes, edges, and shapes.
- The model is then trained and validated, followed by performance evaluation using accuracy, loss, precision, recall, and confusion matrix. Finally, the trained model processes handwritten character images, predicts the recognized characters, and converts them into digital text.

Software and Hardware Tools

- The project can be implemented using Python with TensorFlow/Keras for developing and training the CNN, NumPy for numerical operations, Matplotlib for visualization, and Scikit-learn for performance evaluation. Google Colab or Jupyter Notebook can be used as the development environment.
- A standard computer with an Intel Core i3/i5 or equivalent processor, at least 4 GB RAM, and sufficient storage is suitable for development. A GPU is optional and can be used to accelerate model training.

Microsoft TrOCR for Handwritten Text Extraction

- Used Microsoft TrOCR for handwritten word and text extraction.
- Uses a Vision Encoder–Decoder Transformer model.
- Extracts complete text sequences instead of single characters.
- Improves word ordering and continuous handwriting recognition.
- Integrated with EMNIST CNN for improved text recognition.
- Extracted text is converted into editable digital notes and PDF.

THANK YOU